

Electrical Load List - all consumers by voltage level and bus

4.16 kV MOTORS - SWG-102A

#	Tag	Description	Volt	Starter	Bus	Rated kW	Abs. kW	LF	Eff	PF	Input kW	kvar	kVA	Duty	C kW	I kW	S kW
1	P-101A	Crude charge pump A	4160	VCB-DOL	SWG-102A	710.0	616.7	0.87	0.965	0.89	639.1	327.4	718.0	Continuous	639.1	0.0	0.0
3	P-102A	Desalted crude booster pump A	4160	VCB-DOL	SWG-102A	710.0	615.9	0.87	0.965	0.89	638.2	327.0	717.1	Standby	0.0	0.0	638.2
5	P-112A	Atm. residue / vacuum heater charge pump A	4160	VCB-DOL	SWG-102A	315.0	267.5	0.85	0.955	0.87	280.1	158.8	322.0	Standby	0.0	0.0	280.1
7	P-202A	HVGO pumparound / product pump A	4160	VCB-DOL	SWG-102A	200.0	154.8	0.77	0.955	0.87	162.1	91.9	186.3	Continuous	162.1	0.0	0.0
9	K-102A	H-101 induced-draft fan A	4160	VFD	SWG-102A	200.0	170.1	0.85	0.955	0.95	183.6	60.4	193.3	Continuous	183.6	0.0	0.0
		SWG-102A totals; MD = C + 0.3 I + 0.1 S = 1,077 kW / 528 kvar / 1,199 kVA (PF 0.90)									1,903	965			985	0	918

4.16 kV MOTORS - SWG-102B

#	Tag	Description	Volt	Starter	Bus	Rated kW	Abs. kW	LF	Eff	PF	Input kW	kvar	kVA	Duty	C kW	I kW	S kW
2	P-101B	Crude charge pump B	4160	VCB-DOL	SWG-102B	710.0	616.7	0.87	0.965	0.89	639.1	327.4	718.0	Standby	0.0	0.0	639.1
4	P-102B	Desalted crude booster pump B	4160	VCB-DOL	SWG-102B	710.0	615.9	0.87	0.965	0.89	638.2	327.0	717.1	Continuous	638.2	0.0	0.0
6	P-112B	Atm. residue / vacuum heater charge pump B	4160	VCB-DOL	SWG-102B	315.0	267.5	0.85	0.955	0.87	280.1	158.8	322.0	Continuous	280.1	0.0	0.0
8	P-202B	HVGO pumparound / product pump B	4160	VCB-DOL	SWG-102B	200.0	154.8	0.77	0.955	0.87	162.1	91.9	186.3	Standby	0.0	0.0	162.1
10	K-102B	H-101 induced-draft fan B	4160	VFD	SWG-102B	200.0	170.1	0.85	0.955	0.95	183.6	60.4	193.3	Standby	0.0	0.0	183.6
		SWG-102B totals; MD = C + 0.3 I + 0.1 S = 1,017 kW / 534 kvar / 1,148 kVA (PF 0.89)									1,903	965			918	0	985

480 V / 208 V CONSUMERS - MCC-101A

#	Tag	Description	Volt	Starter	Bus	Rated kW	Abs. kW	LF	Eff	PF	Input kW	kvar	kVA	Duty	C kW	I kW	S kW
11	P-103A	Atm. reflux pump A	480	DOL	MCC-101A	55.0	46.0	0.84	0.935	0.86	49.2	29.2	57.2	Standby	0.0	0.0	49.2
13	P-104A	Unstabilised naphtha pump A	480	DOL	MCC-101A	160.0	118.6	0.74	0.950	0.87	124.8	70.7	143.5	Continuous	124.8	0.0	0.0
15	P-105A	Overhead sour water pump A	480	DOL	MCC-101A	5.5	3.4	0.62	0.875	0.82	3.9	2.7	4.8	Continuous	3.9	0.0	0.0
17	P-106A	TPA pumparound pump A	480	DOL	MCC-101A	110.0	94.0	0.85	0.950	0.87	99.0	56.1	113.8	Standby	0.0	0.0	99.0
19	P-107A	MPA pumparound pump A	480	DOL	MCC-101A	160.0	122.8	0.77	0.950	0.87	129.2	73.2	148.6	Standby	0.0	0.0	129.2
21	P-108A	BPA pumparound pump A	480	DOL	MCC-101A	132.0	101.8	0.77	0.950	0.87	107.1	60.7	123.1	Continuous	107.1	0.0	0.0
23	P-109A	Kero product pump A	480	DOL	MCC-101A	55.0	46.0	0.84	0.935	0.86	49.2	29.2	57.2	Continuous	49.2	0.0	0.0
25	P-110A	Diesel product pump A	480	DOL	MCC-101A	75.0	56.5	0.75	0.935	0.86	60.4	35.8	70.3	Continuous	60.4	0.0	0.0
27	P-111A	Ago product pump A	480	DOL	MCC-101A	45.0	32.4	0.72	0.935	0.86	34.6	20.5	40.2	Continuous	34.6	0.0	0.0
29	P-114A	Desalter wash water pump A	480	DOL	MCC-101A	18.5	13.2	0.71	0.910	0.85	14.5	9.0	17.1	Standby	0.0	0.0	14.5
31	P-115A	Stabiliser reflux / LPG pump A	480	DOL	MCC-101A	18.5	13.9	0.75	0.910	0.85	15.3	9.5	17.9	Standby	0.0	0.0	15.3
33	P-116A	Splitter reflux / LN pump A	480	DOL	MCC-101A	55.0	42.6	0.77	0.935	0.86	45.6	27.0	53.0	Standby	0.0	0.0	45.6
35	P-117A	Heavy naphtha product pump A	480	DOL	MCC-101A	45.0	35.6	0.79	0.935	0.86	38.1	22.6	44.3	Continuous	38.1	0.0	0.0
38	P-201A	LVGO pumparound / product pump A	480	DOL	MCC-101A	110.0	81.3	0.74	0.950	0.87	85.5	48.5	98.3	Continuous	85.5	0.0	0.0
40	P-203A	Slop wax pump A	480	DOL	MCC-101A	5.5	4.7	0.85	0.875	0.82	5.3	3.7	6.5	Standby	0.0	0.0	5.3
42	P-204A	Vacuum residue (incl. quench) pump A	480	DOL	MCC-101A	132.0	112.3	0.85	0.950	0.87	118.2	67.0	135.8	Standby	0.0	0.0	118.2
44	P-205A	Hotwell sour water pump A	480	DOL	MCC-101A	3.7	3.2	0.87	0.875	0.82	3.7	2.6	4.5	Standby	0.0	0.0	3.7
46	P-206A	Hotwell slop oil pump A	480	DOL	MCC-101A	0.8	0.2	0.22	0.800	0.76	0.2	0.2	0.3	Continuous	0.2	0.0	0.0
48	K-101A	H-101 forced-draft fan A	480	DOL	MCC-101A	90.0	73.7	0.82	0.935	0.86	78.8	46.8	91.7	Standby	0.0	0.0	78.8
50	A-101-M1	Atm. overhead condenser - fan motor 1/6 (VFD)	480	VFD	MCC-101A	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
52	A-101-M3	Atm. overhead condenser - fan motor 3/6 (VFD)	480	VFD	MCC-101A	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
54	A-101-M5	Atm. overhead condenser - fan motor 5/6 (VFD)	480	VFD	MCC-101A	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
56	A-103-M1	Kero cooler - fan motor 1/2	480	DOL	MCC-101A	11.0	7.5	0.68	0.910	0.85	8.2	5.1	9.7	Continuous	8.2	0.0	0.0
58	A-104-M1	Diesel cooler - fan motor 1/2	480	DOL	MCC-101A	18.5	14.9	0.81	0.910	0.85	16.4	10.1	19.3	Continuous	16.4	0.0	0.0
60	A-105-M1	Ago cooler - fan motor 1/2	480	DOL	MCC-101A	30.0	24.2	0.81	0.910	0.85	26.6	16.5	31.3	Continuous	26.6	0.0	0.0
62	A-106-M1	Stabiliser overhead condenser - fan motor 1/2	480	DOL	MCC-101A	11.0	8.6	0.78	0.910	0.85	9.4	5.8	11.1	Continuous	9.4	0.0	0.0
64	A-107-M1	Splitter overhead condenser - fan motor 1/4	480	DOL	MCC-101A	30.0	19.3	0.64	0.910	0.85	21.2	13.1	24.9	Continuous	21.2	0.0	0.0
66	A-107-M3	Splitter overhead condenser - fan motor 3/4	480	DOL	MCC-101A	30.0	19.3	0.64	0.910	0.85	21.2	13.1	24.9	Continuous	21.2	0.0	0.0
68	A-108-M1	Heavy naphtha product cooler - fan motor 1/2	480	DOL	MCC-101A	30.0	23.2	0.77	0.910	0.85	25.5	15.8	30.0	Continuous	25.5	0.0	0.0
70	A-201-M1	Lvgo cooler - fan motor 1/2	480	DOL	MCC-101A	15.0	12.4	0.83	0.910	0.85	13.6	8.4	16.0	Continuous	13.6	0.0	0.0
72	A-202-M1	Hvgo Product cooler - fan motor 1/2	480	DOL	MCC-101A	37.0	26.6	0.72	0.935	0.86	28.4	16.9	33.0	Continuous	28.4	0.0	0.0
74	X-101-PA	Demulsifier injection package - metering pump A	480	DOL	MCC-101A	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Continuous	0.6	0.0	0.0
76	X-102-PA	Caustic injection package (desalted crude) - metering pump A	480	DOL	MCC-101A	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Continuous	0.6	0.0	0.0
78	X-103-PA	Neutraliser / filming amine package (atm OH) - metering pump A	480	DOL	MCC-101A	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Continuous	0.6	0.0	0.0
80	X-104-PA	Corrosion inhibitor package (VDU OH) - metering pump A	480	DOL	MCC-101A	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Continuous	0.6	0.0	0.0
82	DT-101A1	D-101A 1st stage - HV transformer 1/2 (150 kVA, 480 V / 13-23 kV)	480	Feeder	MCC-101A	127.5	44.6	0.35	1.000	0.85	44.6	27.7	52.5	Continuous	44.6	0.0	0.0
84	DT-101B1	D-101B 2nd stage - HV transformer 1/2 (150 kVA, 480 V / 13-23 kV)	480	Feeder	MCC-101A	127.5	44.6	0.35	1.000	0.85	44.6	27.7	52.5	Continuous	44.6	0.0	0.0
86	MOV-A	MOV / electric valve actuators - allowance (18 x 1.5 kW), bus A	480	Feeder	MCC-101A	27.0	27.0	1.00	1.000	0.75	27.0	23.8	36.0	Intermittent	0.0	27.0	0.0
88	HTP-101	Electric heat tracing panel - CDU slop & HN lines (allowance)	480	Feeder	MCC-101A	30.0	21.0	0.70	1.000	1.00	21.0	0.0	21.0	Continuous	21.0	0.0	0.0
91	BEF-101A	SS-100 battery-room exhaust fan A	480	Feeder	MCC-101A	1.1	1.1	1.00	1.000	0.80	1.1	0.8	1.4	Continuous	1.1	0.0	0.0
93	HVAC-101A	SS-100 HVAC package A (2 x 100 %)	480	Feeder	MCC-101A	60.0	48.0	0.80	1.000	0.85	48.0	29.7	56.5	Continuous	48.0	0.0	0.0
95	HVAC-102A	FAR-100 HVAC package A (2 x 100 %)	480	Feeder	MCC-101A	25.0	20.0	0.80	1.000	0.85	20.0	12.4	23.5	Continuous	20.0	0.0	0.0
97	CR-A	Field convenience receptacles (120 V), via LTR-101A	208	Fdr/LTR	MCC-101A	10.0	10.0	1.00	1.000	0.90	10.0	4.8	11.1	Intermittent	0.0	10.0	0.0
99	LTG-A	Area & structure lighting (LED), via LTR-101A 480-208Y/120 V	208	Fdr/LTR	MCC-101A	24.0	24.0	1.00	1.000	0.95	24.0	7.9	25.3	Continuous	24.0	0.0	0.0
101	LTG-BA	Building lighting SS-100 / FAR-100, via LTR-101A	208	Fdr/LTR	MCC-101A	5.0	5.0	1.00	1.000	0.95	5.0	1.6	5.3	Continuous	5.0	0.0	0.0
103	SP-A	Building small power & receptacles, via LTR-101A	208	Fdr/LTR	MCC-101A	10.0	6.0	0.60	1.000	0.90	6.0	2.9	6.7	Continuous	6.0	0.0	0.0
105	WR-A	Welding receptacles 3 x 63 A, 480 V, bus A	480	Feeder	MCC-101A	75.0	75.0	1.00	1.000	0.70	75.0	76.5	107.1	Intermittent	0.0	75.0	0.0
107	CP-101	Cathodic protection transformer-rectifier (buried piping / tank bottoms in plot)	480	Feeder	MCC-101A	8.0	6.4	0.80	1.000	0.80	6.4	4.8	8.0	Continuous	6.4	0.0	0.0

#	Tag	Description	Volt	Starter	Bus	Rated kW	Abs. kW	LF	Eff	PF	Input kW	kvar	kVA	Duty	C kW	I kW	S kW
108	BC-101A	125 V DC battery charger A (switchgear control), 2 x 100 %	480	Feeder	MCC-101A	6.0	3.0	0.50	0.900	0.85	3.3	2.1	3.9	Continuous	3.3	0.0	0.0
110	UPS-101A	UPS-101A rectifier input (2 x 100 % 100 kVA, 50 % share + battery recharge)	480	Feeder	MCC-101A	34.0	34.0	1.00	1.000	0.95	34.0	11.2	35.8	Continuous	34.0	0.0	0.0
112	HST-101	Monorail hoists P-101/P-102 maintenance (2 x 7.5 kW)	480	Feeder	MCC-101A	15.0	15.0	1.00	1.000	0.80	15.0	11.2	18.8	Intermittent	0.0	15.0	0.0
		MCC-101A totals; MD = C + 0.3 I + 0.1 S = 1,178 kW / 643 kvar / 1,342 kVA (PF 0.88)									1,769	1,016			1,084	127	559

480 V / 208 V CONSUMERS - MCC-101B

#	Tag	Description	Volt	Starter	Bus	Rated kW	Abs. kW	LF	Eff	PF	Input kW	kvar	kVA	Duty	C kW	I kW	S kW
12	P-103B	Atm. reflux pump B	480	DOL	MCC-101B	55.0	46.0	0.84	0.935	0.86	49.2	29.2	57.2	Continuous	49.2	0.0	0.0
14	P-104B	Unstabilised naphtha pump B	480	DOL	MCC-101B	160.0	118.6	0.74	0.950	0.87	124.8	70.7	143.5	Standby	0.0	0.0	124.8
16	P-105B	Overhead sour water pump B	480	DOL	MCC-101B	5.5	3.4	0.62	0.875	0.82	3.9	2.7	4.8	Standby	0.0	0.0	3.9
18	P-106B	TPA pumparound pump B	480	DOL	MCC-101B	110.0	94.0	0.85	0.950	0.87	99.0	56.1	113.8	Continuous	99.0	0.0	0.0
20	P-107B	MPA pumparound pump B	480	DOL	MCC-101B	160.0	122.8	0.77	0.950	0.87	129.2	73.2	148.6	Continuous	129.2	0.0	0.0
22	P-108B	BPA pumparound pump B	480	DOL	MCC-101B	132.0	101.8	0.77	0.950	0.87	107.1	60.7	123.1	Standby	0.0	0.0	107.1
24	P-109B	Kero product pump B	480	DOL	MCC-101B	55.0	46.0	0.84	0.935	0.86	49.2	29.2	57.2	Standby	0.0	0.0	49.2
26	P-110B	Diesel product pump B	480	DOL	MCC-101B	75.0	56.5	0.75	0.935	0.86	60.4	35.8	70.3	Standby	0.0	0.0	60.4
28	P-111B	Ago product pump B	480	DOL	MCC-101B	45.0	32.4	0.72	0.935	0.86	34.6	20.5	40.2	Standby	0.0	0.0	34.6
30	P-114B	Desalter wash water pump B	480	DOL	MCC-101B	18.5	13.2	0.71	0.910	0.85	14.5	9.0	17.1	Continuous	14.5	0.0	0.0
32	P-115B	Stabiliser reflux / LPG pump B	480	DOL	MCC-101B	18.5	13.9	0.75	0.910	0.85	15.3	9.5	17.9	Continuous	15.3	0.0	0.0
34	P-116B	Splitter reflux / LN pump B	480	DOL	MCC-101B	55.0	42.6	0.77	0.935	0.86	45.6	27.0	53.0	Continuous	45.6	0.0	0.0
36	P-117B	Heavy naphtha product pump B	480	DOL	MCC-101B	45.0	35.6	0.79	0.935	0.86	38.1	22.6	44.3	Standby	0.0	0.0	38.1
37	P-118	Desalter mud-wash / recycle (single, intermittent)	480	DOL	MCC-101B	7.5	5.4	0.71	0.875	0.82	6.1	4.3	7.5	Intermittent	0.0	6.1	0.0
39	P-201B	LVGO pumparound / product pump B	480	DOL	MCC-101B	110.0	81.3	0.74	0.950	0.87	85.5	48.5	98.3	Standby	0.0	0.0	85.5
41	P-203B	Slop wax pump B	480	DOL	MCC-101B	5.5	4.7	0.85	0.875	0.82	5.3	3.7	6.5	Continuous	5.3	0.0	0.0
43	P-204B	Vacuum residue (incl. quench) pump B	480	DOL	MCC-101B	132.0	112.3	0.85	0.950	0.87	118.2	67.0	135.8	Continuous	118.2	0.0	0.0
45	P-205B	Hotwell sour water pump B	480	DOL	MCC-101B	3.7	3.2	0.87	0.875	0.82	3.7	2.6	4.5	Continuous	3.7	0.0	0.0
47	P-206B	Hotwell slop oil pump B	480	DOL	MCC-101B	0.8	0.2	0.22	0.800	0.76	0.2	0.2	0.3	Standby	0.0	0.0	0.2
49	K-101B	H-101 forced-draft fan B	480	DOL	MCC-101B	90.0	73.7	0.82	0.935	0.86	78.8	46.8	91.7	Continuous	78.8	0.0	0.0
51	A-101-M2	Atm. overhead condenser - fan motor 2/6 (VFD)	480	VFD	MCC-101B	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
53	A-101-M4	Atm. overhead condenser - fan motor 4/6 (VFD)	480	VFD	MCC-101B	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
55	A-101-M6	Atm. overhead condenser - fan motor 6/6 (VFD)	480	VFD	MCC-101B	55.0	45.0	0.82	0.935	0.95	49.6	16.3	52.2	Continuous	49.6	0.0	0.0
57	A-103-M2	Kero cooler - fan motor 2/2	480	DOL	MCC-101B	11.0	7.5	0.68	0.910	0.85	8.2	5.1	9.7	Continuous	8.2	0.0	0.0
59	A-104-M2	Diesel cooler - fan motor 2/2	480	DOL	MCC-101B	18.5	14.9	0.81	0.910	0.85	16.4	10.1	19.3	Continuous	16.4	0.0	0.0
61	A-105-M2	Ago cooler - fan motor 2/2	480	DOL	MCC-101B	30.0	24.2	0.81	0.910	0.85	26.6	16.5	31.3	Continuous	26.6	0.0	0.0
63	A-106-M2	Stabiliser overhead condenser - fan motor 2/2	480	DOL	MCC-101B	11.0	8.6	0.78	0.910	0.85	9.4	5.8	11.1	Continuous	9.4	0.0	0.0
65	A-107-M2	Splitter overhead condenser - fan motor 2/4	480	DOL	MCC-101B	30.0	19.3	0.64	0.910	0.85	21.2	13.1	24.9	Continuous	21.2	0.0	0.0
67	A-107-M4	Splitter overhead condenser - fan motor 4/4	480	DOL	MCC-101B	30.0	19.3	0.64	0.910	0.85	21.2	13.1	24.9	Continuous	21.2	0.0	0.0
69	A-108-M2	Heavy naphtha product cooler - fan motor 2/2	480	DOL	MCC-101B	30.0	23.2	0.77	0.910	0.85	25.5	15.8	30.0	Continuous	25.5	0.0	0.0
71	A-201-M2	Lvgo cooler - fan motor 2/2	480	DOL	MCC-101B	15.0	12.4	0.83	0.910	0.85	13.6	8.4	16.0	Continuous	13.6	0.0	0.0
73	A-202-M2	Hvgo Product cooler - fan motor 2/2	480	DOL	MCC-101B	37.0	26.6	0.72	0.935	0.86	28.4	16.9	33.0	Continuous	28.4	0.0	0.0
75	X-101-PB	Demulsifier injection package - metering pump B	480	DOL	MCC-101B	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Standby	0.0	0.0	0.6
77	X-102-PB	Caustic injection package (desalted crude) - metering pump B	480	DOL	MCC-101B	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Standby	0.0	0.0	0.6
79	X-103-PB	Neutraliser / filming amine package (atm OH) - metering pump B	480	DOL	MCC-101B	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Standby	0.0	0.0	0.6
81	X-104-PB	Corrosion inhibitor package (VDU OH) - metering pump B	480	DOL	MCC-101B	0.8	0.4	0.60	0.800	0.76	0.6	0.5	0.7	Standby	0.0	0.0	0.6
83	DT-101A2	D-101A 1st stage - HV transformer 2/2 (150 kVA, 480 V / 13-23 kV)	480	Feeder	MCC-101B	127.5	44.6	0.35	1.000	0.85	44.6	27.7	52.5	Continuous	44.6	0.0	0.0
85	DT-101B2	D-101B 2nd stage - HV transformer 2/2 (150 kVA, 480 V / 13-23 kV)	480	Feeder	MCC-101B	127.5	44.6	0.35	1.000	0.85	44.6	27.7	52.5	Continuous	44.6	0.0	0.0
87	MOV-B	MOV / electric valve actuators - allowance (18 x 1.5 kW), bus B	480	Feeder	MCC-101B	27.0	27.0	1.00	1.000	0.75	27.0	23.8	36.0	Intermittent	0.0	27.0	0.0
89	HTP-102	Electric heat tracing panel - VDU slop wax / slop oil lines (allowance)	480	Feeder	MCC-101B	40.0	28.0	0.70	1.000	1.00	28.0	0.0	28.0	Continuous	28.0	0.0	0.0
90	AH-101	Analyser house AH-101 HVAC & utilities (analysers on UPS)	480	Feeder	MCC-101B	15.0	12.0	0.80	1.000	0.85	12.0	7.4	14.1	Continuous	12.0	0.0	0.0
92	BEF-101B	SS-100 battery-room exhaust fan B	480	Feeder	MCC-101B	1.1	1.1	1.00	1.000	0.80	1.1	0.8	1.4	Standby	0.0	0.0	1.1
94	HVAC-101B	SS-100 HVAC package B (2 x 100 %)	480	Feeder	MCC-101B	60.0	48.0	0.80	1.000	0.85	48.0	29.7	56.5	Standby	0.0	0.0	48.0
96	HVAC-102B	FAR-100 HVAC package B (2 x 100 %)	480	Feeder	MCC-101B	25.0	20.0	0.80	1.000	0.85	20.0	12.4	23.5	Standby	0.0	0.0	20.0
98	CR-B	Field convenience receptacles (120 V), via LTR-101B	208	Fdr/LTR	MCC-101B	10.0	10.0	1.00	1.000	0.90	10.0	4.8	11.1	Intermittent	0.0	10.0	0.0
100	LTG-B	Area & structure lighting (LED), via LTR-101B 480-208Y/120 V	208	Fdr/LTR	MCC-101B	24.0	24.0	1.00	1.000	0.95	24.0	7.9	25.3	Continuous	24.0	0.0	0.0
102	LTG-BB	Building lighting SS-100 / FAR-100, via LTR-101B	208	Fdr/LTR	MCC-101B	5.0	5.0	1.00	1.000	0.95	5.0	1.6	5.3	Continuous	5.0	0.0	0.0
104	SP-B	Building small power & receptacles, via LTR-101B	208	Fdr/LTR	MCC-101B	10.0	6.0	0.60	1.000	0.90	6.0	2.9	6.7	Continuous	6.0	0.0	0.0
106	WR-B	Welding receptacles 3 x 63 A, 480 V, bus B	480	Feeder	MCC-101B	75.0	75.0	1.00	1.000	0.70	75.0	76.5	107.1	Intermittent	0.0	75.0	0.0
109	BC-101B	125 V DC battery charger B (switchgear control), 2 x 100 %	480	Feeder	MCC-101B	6.0	3.0	0.50	0.900	0.85	3.3	2.1	3.9	Continuous	3.3	0.0	0.0
111	UPS-101B	UPS-101B rectifier input (2 x 100 % 100 kVA, 50 % share + battery recharge)	480	Feeder	MCC-101B	34.0	34.0	1.00	1.000	0.95	34.0	11.2	35.8	Continuous	34.0	0.0	0.0
113	SB-101	H-101 convection sootblowers (8 x 0.75 kW, electric retract)	480	Feeder	MCC-101B	6.0	6.0	1.00	1.000	0.80	6.0	4.5	7.5	Intermittent	0.0	6.0	0.0
		MCC-101B totals; MD = C + 0.3 I + 0.1 S = 1,175 kW / 634 kvar / 1,335 kVA (PF 0.88)									1,779	1,016			1,080	124	575

Maximum demand summary and transformer loading

Bus	kV	No.	Conn. kW	C kW	I kW	S kW	MD kW	MD kvar	MD kVA	PF
MCC-101A	0.48	51	1,769	1,084	127	559	1,178	643	1,342	0.88
MCC-101B	0.48	52	1,779	1,080	124	575	1,175	634	1,335	0.88
SWG-102A	4.16	5	1,903	985	0	918	1,077	528	1,199	0.90
SWG-102B	4.16	5	1,903	918	0	985	1,017	534	1,148	0.89
SWG-102A incl. LV through-load	4.16						2,266	1,197	2,563	0.88
SWG-102B incl. LV through-load	4.16						2,203	1,193	2,505	0.88
SWG-101A	13.8						2,284	1,269	2,613	0.87
SWG-101B	13.8						2,221	1,265	2,556	0.87
LV total (MCC-101A+B)	0.48	103	3,548	2,163	251	1,134	2,352	1,277	2,676	0.88
MV motors total (SWG-102A+B)	4.16	10	3,806	1,903	0	1,903	2,093	1,062	2,347	0.89
UNIT TOTAL at 13.8 kV (incl. TR losses)	13.8	113					4,505	2,533	5,168	0.87

TR-101 / TR-102: 13.8/4.16 kV, 7,500 / 9,375 kVA ONAN/ONAF, Z 7.0 %. Contingency MD (both sections on one transformer) 5,068 kVA = 68 % of ONAN; MD x 1.25 = 6,335 kVA = 68 % of ONAF.
TR-103 / TR-104: 4.16/0.48 kV, 3,000 / 3,750 kVA ONAN/ONAF, Z 5.75 %. Contingency MD (both sections on one transformer) 2,676 kVA = 89 % of ONAN; MD x 1.25 = 3,345 kVA = 89 % of ONAF.

UPS consumers (120 V, via UPS-101A/B; not included in bus totals above except as UPS rectifier input)

Code	Description	kVA total	Feed A kVA	Feed B kVA
UPS-DCS	Distributed control system (controllers, I/O, servers, HMI in FAR)	22	11	11
UPS-SIS	Safety instrumented system incl. H-101/H-201 BMS, ESD solenoids	14	7	7
UPS-FGS	Fire & gas system incl. detectors/beacons	6	3	3
UPS-TEL	Telecom: PAGA, CCTV, LAN/fibre, radio	8	4	4
UPS-ANZ	Process analysers (AH-101)	6	3	3
UPS-MISC	Misc. instrument panels, MCC/IED networking, metering	4	2	2

Basis

- Sources: motor ratings, absorbed power, fan counts and package data from data/equipment.json (process/mechanical sizing). Absorbed power used where available; 'Input kW' = absorbed / efficiency (/ VFD efficiency 0.97 where applicable).
- Voltage levels: motors >= 200 kW at 4.16 kV (VCB, DOL unless VFD); motors < 200 kW at 480 V (MCC, DOL). VFDs: A-101 overhead condenser fans (4 x 90 kW, process temperature control and energy) and H-101 ID fans K-102A/B (draft control, 4.16 kV MV drive).
- Efficiency / PF by motor size (NEMA Premium / IEEE 841 typical): <=1.5 kW 0.80/0.76; <=7.5 kW 0.875/0.82; <=30 kW 0.91/0.85; <=90 kW 0.935/0.86; <=160 kW 0.95/0.87; LV >160 kW 0.955/0.87; MV <500 kW 0.955/0.87; MV >=500 kW 0.965/0.89. VFD supply PF 0.95.
- Duty: C = continuous, I = intermittent, S = standby. Pump pairs A/B (2 x 100 %): A always on bus A, B on bus B; the normally running unit of each pair (C) is chosen to balance the two bus sections, the other is standby (S). Air-cooler fans all continuous, split alternately across the buses.
- Maximum demand (IEEE 141 / IEC style diversity): MD = 1.0 C + 0.3 I + 0.1 S, applied to kW and kvar separately; kVA by vector sum.
- Allowances (no process data available - FEED assumptions, to be replaced by vendor data): MOV/actuators 36 x 1.5 kW; heat tracing 70 kW (slop/HN/slop-wax lines); lighting 2 x 29 kW LED + building lighting; small power/receptacles; welding receptacles 6 x 63 A; SS-100 HVAC 2 x 60 kW, FAR-100 HVAC 2 x 25 kW; cathodic protection 8 kW; analyser house 15 kW; UPS loads 60 kVA; 125 V DC chargers 2 x 6 kW.
- Desalter transformers: 2 x 150 kVA per stage (D-101A, D-101B), 480 V primary, load factor 0.35 (vendor to confirm), PF 0.85; one transformer of each stage on each bus.
- Chemical injection packages X-101..X-104: 2 x 100 % metering pumps 0.75 kW each, absorbed 60 %.
- 208 V loads are supplied via lighting transformers LTR-101A/B (480-208Y/120 V) from MCC-101A/B.
- UPS consumers (120 V) are listed separately; they are supplied through UPS-101A/B whose rectifier input (incl. battery recharge) appears on MCC-101A/B. Not double counted.